

CNN Architectures: From LeNet to ResNet

Scaler School of Technology | Deep Learning I | Week 4 — Session 2/2 | May 2026



What Is This Lecture About?

Last session you learned the convolution operation itself. This session traces how people **stacked those operations into full architectures** — and why each design was a breakthrough. You'll see how CNN architectures evolved from 5-layer digit readers to 152-layer superhuman image classifiers in just 17 years.

1 LeNet to AlexNet — Scale Changes Everything

The first CNN (1998) recognized digits with 60K parameters. AlexNet (2012) scaled the same idea to 60M parameters with ReLU, dropout, and GPUs — and kicked off the deep learning revolution. *Like going from a home kitchen to an industrial factory — same recipe, vastly bigger scale.*

2 VGG and Inception — Depth vs. Efficiency

VGG proved that stacking tiny 3×3 filters deep beats large filters (fewer params, more non-linearity). Inception took a different path: process multiple scales in parallel using 1×1 bottlenecks, achieving better accuracy with $28 \times$ fewer parameters than VGG. *Like choosing between a long assembly line vs. a team of specialized experts working in parallel.*

3 ResNet — Skip Connections Solve the Depth Problem

Plain networks degrade when made deeper (paradoxically getting worse). ResNet's skip connections let gradients flow directly through the network, enabling 152 layers and superhuman accuracy. This is the single most important architectural idea in modern deep learning.

What to Expect in the Session

- Tracing tensor dimensions through LeNet-5
- Parameter counting: why VGG is wasteful
- ★ Building and testing a **residual block** from scratch
- Visualizing gradient flow: plain net vs ResNet
- 2 quizzes on architecture trade-offs
- Architecture comparison dashboard (params, depth, accuracy)

How to Prepare (Optional but Helpful)

- ✓ **Recall** how convolution, padding, and stride affect output dimensions (Session 7)
- ✓ **Recall** vanishing gradients — we'll see how skip connections solve this
- ✓ **Think about:** "If a 56-layer network should be at least as good as a 20-layer one (the extra layers could just learn identity), why might it actually perform worse?"
- ✓ **Skim** `torchvision.models` documentation if curious about pretrained models

Duration: ~2 hours
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Prerequisites: Convolution operation, pooling, parameter sharing (Session 7)

Instructor: